Current Electricity

Electric current — The Rate of directional flow of electric charge is c/d electric current.

gn time 't' → $I_{Avg} = \frac{\{\Delta Q\}}{\Delta t} = \frac{(ne)_{Tranfered}}{\Delta t}$

$$\boxed{I_{Avg} = \frac{\int I_{gnst}\,dt}{\int dt}}$$

→ Ampere (fundamental Physical quantity)

at time (t) → $I_{gnst} = \left(\frac{dQ}{dt}\right)$ = The rate of dir^n flow of charge w.r.t time

I = slope of Q/time graph

→ unit Amp (C.G.S → Biot)

→ Scalar (Tensor)

→ dir^n along flow of +ve charge

→ does not follow Triangle law of vector add^n

$\Delta Q = \int dq = \int I\,dt$ = Area of I/t graph

Q → I

$\frac{dQ}{dt} = I_{gnst}$ = Slope of Q/t graph

diff^n $= \frac{dI}{dt}$ = slope of I/t graph

Int → given Area

$\boxed{I_{Avg} = \frac{\Delta Q}{\Delta t}}$ ✗